

Netflix Titles — Content Analysis

Project Report — Analytical Questions with Explanation & Insights

Data Analysis & Visualization Project | Dataset: Netflix Titles (2021)

1. Overview

Q1. What is the objective of this project?

To analyze the Netflix titles catalog in order to understand the mix of Movies versus TV Shows, identify the top content-producing countries, track how the catalog has grown over time, study the distribution of content ratings and genres, and derive insights that support content strategy and acquisition decisions.

Q2. Why is this analysis useful?

Netflix is one of the largest global streaming platforms, and understanding its catalog composition reveals content strategy, regional investment patterns, and audience targeting — useful for benchmarking and for anyone studying streaming media trends.

Q3. What does the dataset contain?

Source	Netflix Titles dataset (2021 snapshot)
Format	CSV (structured data)
Rows / Columns	8,807 titles across 12 columns
Key Fields	type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description
Tools Used	Python, Pandas, Matplotlib, Seaborn, Google Colab

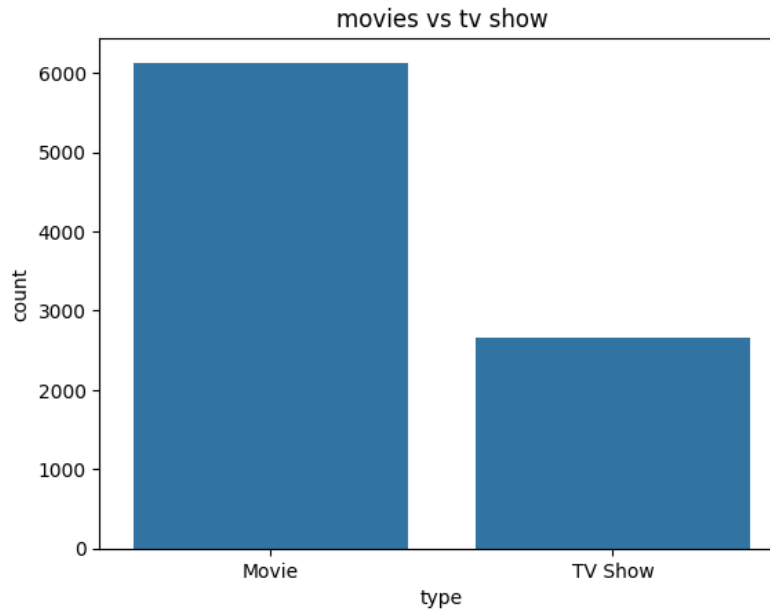
Q4. What steps were followed in this project?

- Data Collection — obtaining the Netflix titles dataset
- Data Cleaning & Preprocessing — handling missing values in director/cast/country, parsing date_added into a proper datetime, creating a year_added feature
- Exploratory Data Analysis (EDA) — understanding distributions of type, country, rating, and genre
- Data Visualization — count plots, bar charts, and line charts comparing key variables
- Insight Generation — interpreting charts to find catalog trends
- Conclusion & Recommendations — summarizing findings for content strategy

2. Data Analysis & Visualization

Q1. What is the split between Movies and TV Shows?

Answer: Movies make up the large majority of the catalog, at roughly 70% of titles, with TV Shows accounting for the remaining 30% (about 6,100 Movies vs. about 2,700 TV Shows).



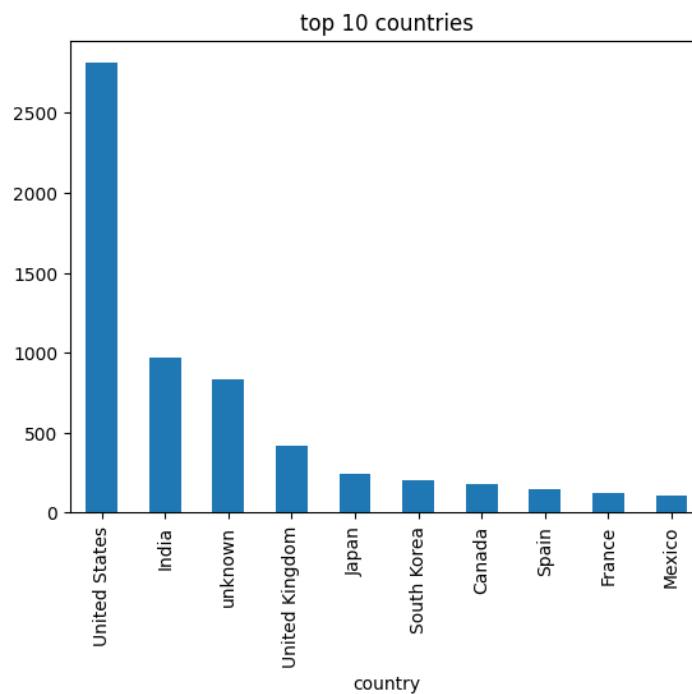
Insights:

- Movies outnumber TV Shows by more than 2 to 1
- This suggests Netflix's catalog has historically been movie-heavy, even as originals investment has grown

📌 **Strategy:** Track whether the Movie:TV ratio is shifting over time, since TV Shows tend to drive longer-term subscriber retention.

Q2. Which countries produce the most Netflix content?

Answer: The United States leads by a wide margin, followed by India, then a large 'unknown' (missing-data) group, and the United Kingdom. Japan, South Korea, Canada, Spain, France, and Mexico round out the top 10.



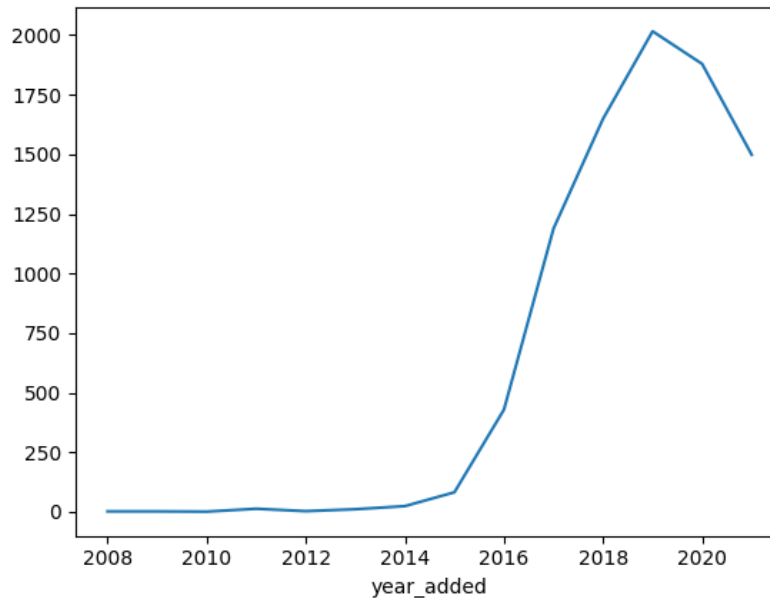
Insights:

- The US and India together dominate content origin
- A sizeable share of titles have no recorded country, which limits precision in geographic analysis

🔗 **Strategy: Improve metadata completeness for the country field, and study India's strong share as a signal for continued regional content investment.**

Q3. How has the number of titles added to Netflix grown over time?

Answer: Additions were minimal before 2015, rose sharply between 2016 and 2019, peaked around 2019, and declined slightly in 2020 and 2021.



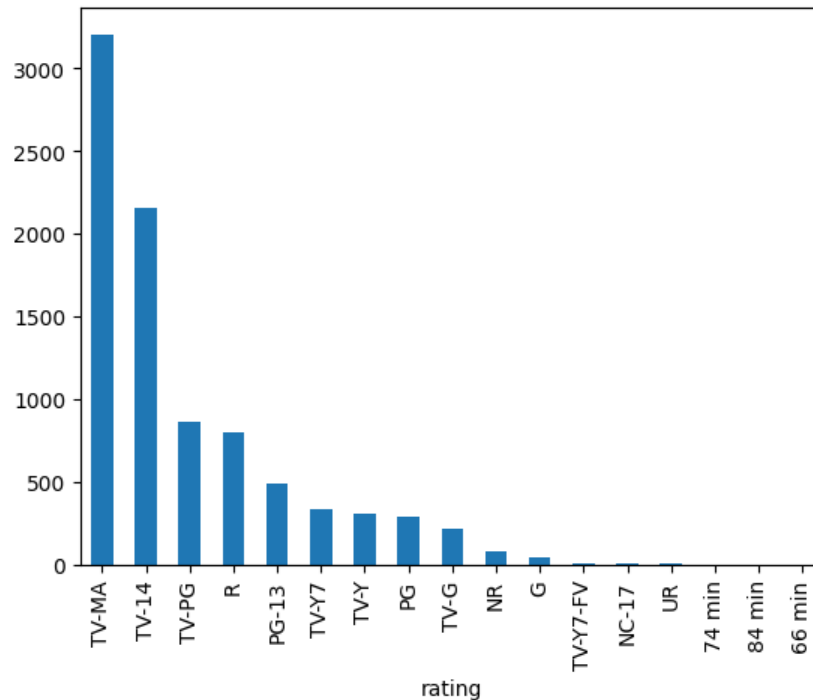
Insights:

- The 2016–2019 window marks Netflix's most aggressive catalog expansion
- The 2020–2021 dip likely reflects production slowdowns during the pandemic

🔗 **Strategy: Use the pre-2020 growth rate as a benchmark, and monitor whether additions recover post-pandemic.**

Q4. What are the most common content ratings?

Answer: TV-MA is the most common rating by a clear margin, followed by TV-14. PG-13, R, and TV-PG make up a secondary tier, with family-oriented ratings (TV-Y, TV-G, PG) further behind.



Insights:

- The catalog skews toward mature-audience content (TV-MA + TV-14 together are the two largest categories)
- Family and kids' ratings form a smaller but still meaningful share

📌 **Strategy: Balance mature-content acquisition with continued investment in family/kids programming to serve a broader subscriber base.**

Q5. What are the most common genres / categories?

Answer: "Dramas, International Movies" is the single largest category, followed closely by Documentaries and Stand-Up Comedy. Several other top categories are combinations involving International Movies, Comedies, and Dramas.

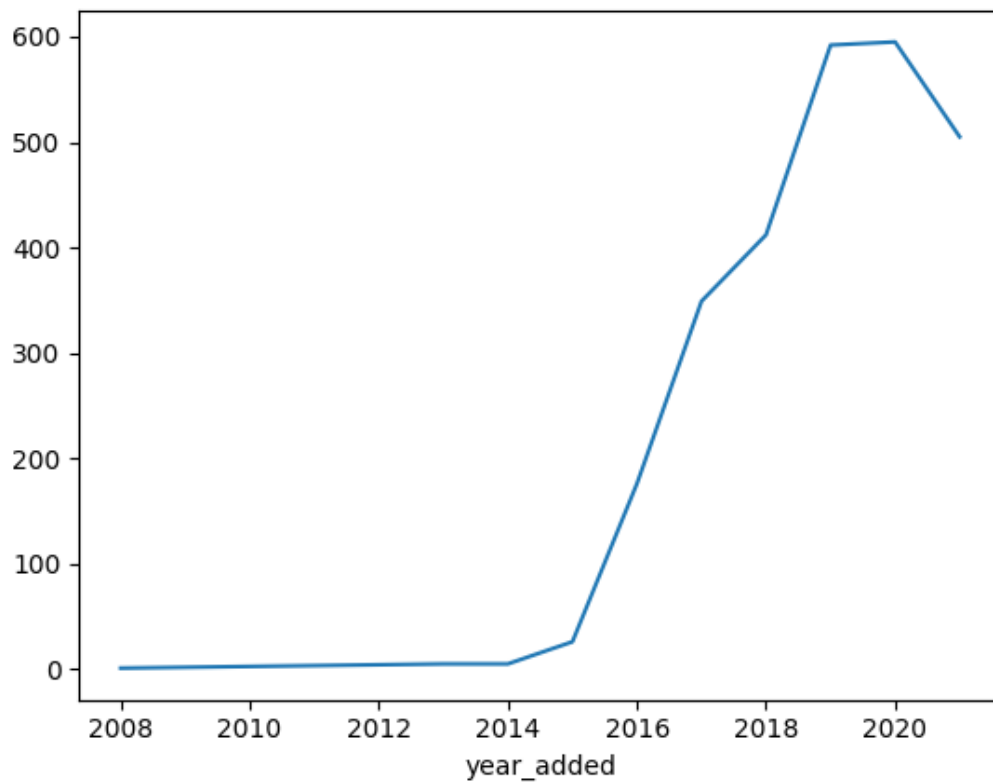
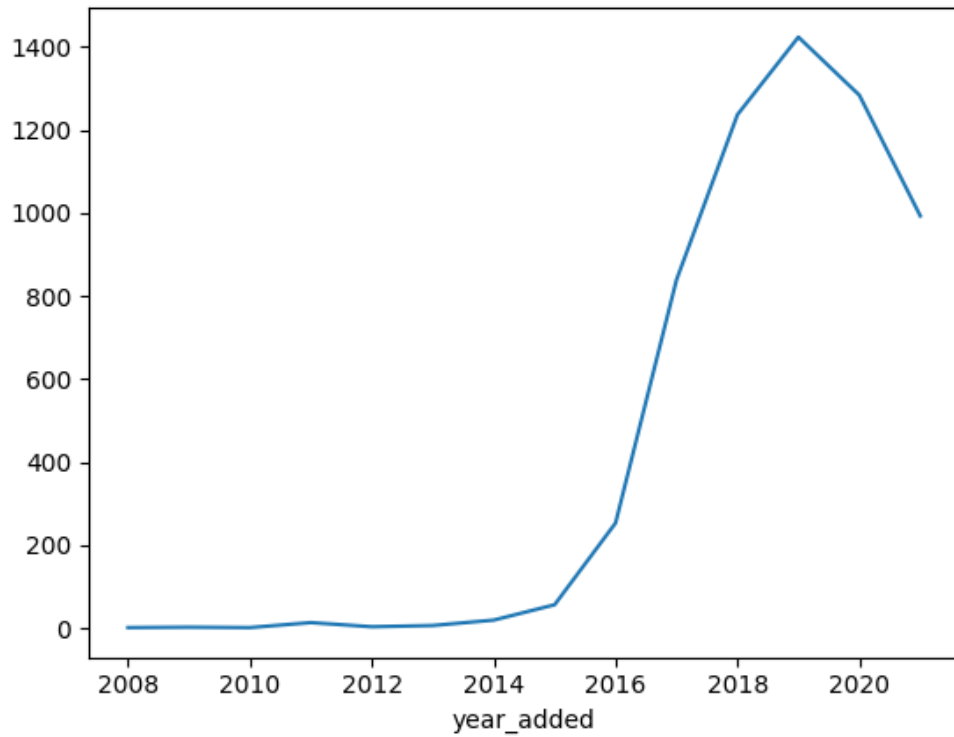
Insights:

- Dramas, International Movies — 362 titles
- Documentaries — 359 titles
- Stand-Up Comedy — 334 titles
- Comedies, Dramas, International Movies — 274 titles
- Dramas, Independent Movies, International Movies — 252 titles
- Kids' TV — 219 titles
- Children & Family Movies — 215 titles

📌 **Strategy: International drama and documentary content are proven high-volume categories worth continued investment.**

Q6. Do Movies and TV Shows follow the same growth pattern?

Answer: Both follow a similar overall shape — flat before 2015, sharp growth from 2016–2019, then a slight decline — but Movies consistently outpace TV Shows in volume added each year.



Insights:

- Movie additions peaked near 1,400 titles in a single year around 2019
- TV Show additions peaked near 600 titles in the same period, roughly a third of the movie peak

🔑 **Strategy: Continue growing the TV Show pipeline, since it lags Movies in volume but tends to build stronger long-term engagement.**

3. Key Insights & Findings

- The catalog is Movie-heavy, at roughly a 70/30 split versus TV Shows
- The United States and India are the dominant content sources
- Catalog growth was concentrated in 2016–2019, with a pandemic-era slowdown in 2020–2021
- Content skews toward mature ratings (TV-MA, TV-14)
- International dramas, documentaries, and stand-up comedy are the leading genres
- Missing metadata (director, cast, country) is common enough to warrant data-quality attention

4. Recommendations

- Continue investing in international drama, documentary, and stand-up comedy content given their proven volume
- Grow the TV Show pipeline to close the gap with Movie additions
- Improve metadata completeness, especially for country and director fields
- Monitor whether content additions recover to pre-2020 levels
- Maintain a mix of mature and family-oriented ratings to serve a broad subscriber base

Final Conclusion

This project analyzed the Netflix titles dataset to uncover key patterns in content type, geography, ratings, genres, and catalog growth over time. Movies dominate the catalog by volume, the US and India are the leading content sources, and 2016–2019 was Netflix's period of most aggressive expansion before a pandemic-era slowdown. These findings highlight how content strategy and catalog composition can be understood through structured data analysis, supporting more informed content acquisition and investment decisions.